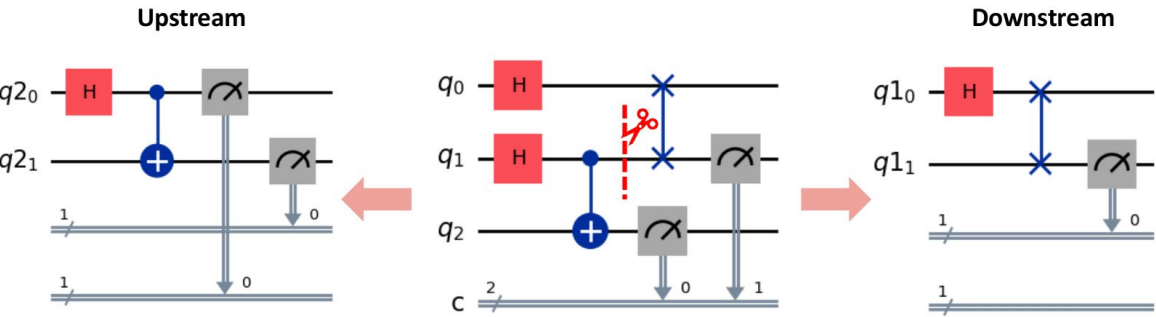


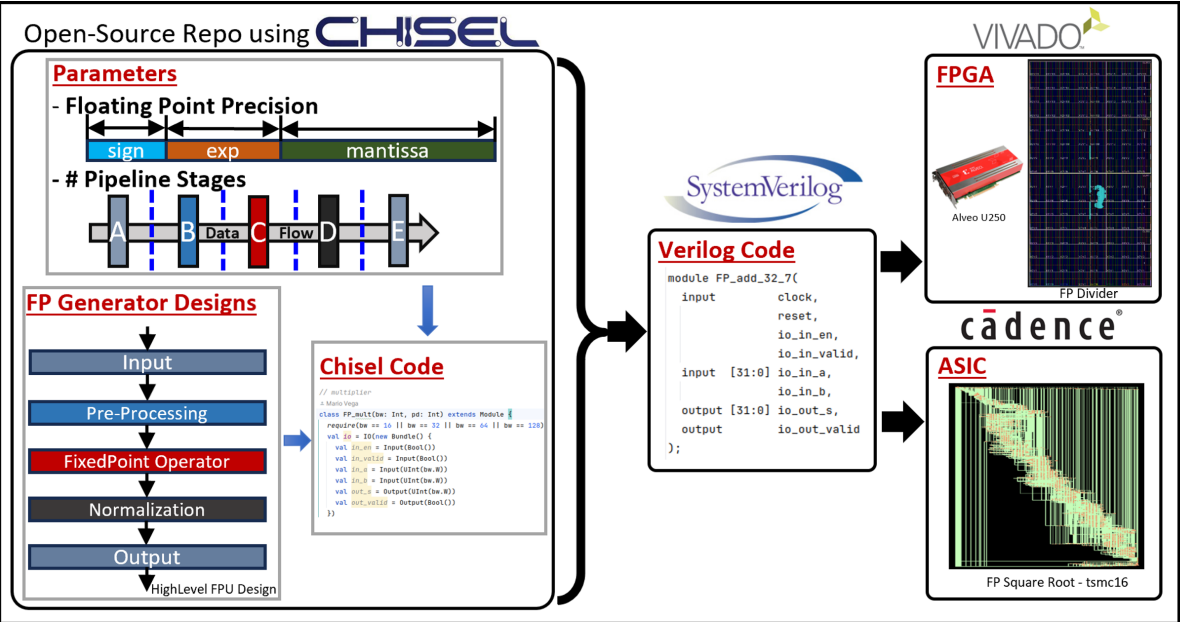
Classical-Quantum Circuit Design and Simulation

Xiaokun Yang, Professor, UH Clear-Lake | Affiliate Faculty, LBNL

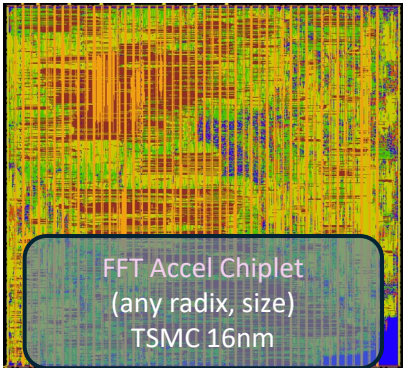
- Scalable Quantum Circuit Simulation via Circuit Cutting [1] and Specialized Hardware Design Acceleration [2-3]



- Beyond Moore's Law: Specialized ASIC Design Acceleration [4-6]



NSF #2216923.
NSF #2243981

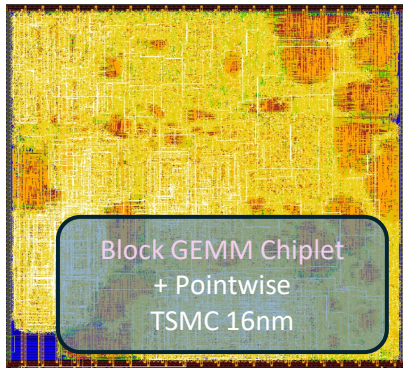


Chiplet in 16nm TSMC [7]

Area eff.: 4.18 TF/mm²
Energy eff.: 4.8 TF/W

NVIDIA H100 in 4nm TSMC (10x denser than 16nm)

Area eff: 0.08 TF/mm²
Energy eff: 0.0957 TF/W



Accomplishment [7]:

Demonstrated accelerator tile generator in 16nm TSMC that outperforms NVIDIA H100 by **50x** in raw performance/area and **50x** in energy efficiency/flop



DOE ASCR 24-3210, Base Research, Thrust E

[1] Q. Liu, X. Yang, et.al, QCE 2026. [2] X. Yang, et. al, HPDC 2026. [3] J. W. Turner, X. Yang, et. al, ICDM 2025. [4] X. Yang, et.al, TCAD 2026. [5] D. T. Popovici, X. Yang, et. al., FPGA 2026. [6] A. Ioannou, X. Yang, el.al., IPDSP 2026. [7] M. Vega, X. Yang, el.al., HPEC 2024.



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